Mounting-element statistics

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Figures are counts, shares and descriptive statistics of fields that are actually stored. A null load, spacing or material stays empty. An element can match more than one clamp, so product-hit totals are not a partition of the element count. A system assignment total can exceed the element count when one element sits in more than one system. Coverage below uses substrate none, so concrete anchors are not counted as matches. The architecture ceilings are gypsum under timber joists.

Headline figures

measure	value
products	23
guidance records	3
manufacturers	6
primary products	18
companion products	5
inventory size groups	98
inventory elements	1163
elements matched on a published size	777
elements matched with an assumption	290
elements unmatched	96
share matched on a published size	66.8%
share matched with an assumption	24.9%
share unmatched	8.3%
pipe segment length, m	303.832
duct segment length, m	118.376
models	3

Products by manufacturer

Each product counts once. Share is the share of the catalogue, not of the building.

manufacturer	products	share	primary	companion	with load	load empty	element types
Hilti	6	26.1%	3	3	3	3	channel, pipe, threaded_rod
Kermi	1	4.3%	1	0	0	1	radiator
Müpro	3	13.0%	3	0	0	3	pipe
Sikla	3	13.0%	3	0	3	0	duct_rectangular, duct_round
Walraven	2	8.7%	2	0	0	2	pipe
fischer	8	34.8%	6	2	7	1	ceiling_anchor, pipe

Products by product line

product line	manufacturer	products	share	roles	element types
FRS Plus	fischer	4	17.4%	primary	pipe
MP-MI	Hilti	3	13.0%	primary	pipe
OPTIMAL	Müpro	3	13.0%	primary	pipe
AM	Hilti	2	8.7%	companion	threaded_rod
EA II	fischer	2	8.7%	companion	ceiling_anchor
LUW	Sikla	2	8.7%	primary	duct_rectangular
BISMAT 2000	Walraven	1	4.3%	primary	pipe
BISMAT KSB 2	Walraven	1	4.3%	primary	pipe
FRS A2	fischer	1	4.3%	primary	pipe
FRS-L Universal	fischer	1	4.3%	primary	pipe
MQ	Hilti	1	4.3%	companion	channel
PLU 2G-N m.E.	Sikla	1	4.3%	primary	duct_round
Wandkonsole variabel	Kermi	1	4.3%	primary	radiator

Products by element type

duct_flex is listed even when the count is zero: the catalogue has no flex-duct hanger. A product with several element types would be counted in each column of the cross table and once in this table per type.

element type	products	share of catalogue	primary	companion	with OD window	with nominal diameter	with load
pipe	14	60.9%	14	0	14	0	8
duct_round	1	4.3%	1	0	0	1	1
duct_rectangular	2	8.7%	2	0	0	0	2
duct_flex	0	0.0%	0	0	0	0	0
radiator	1	4.3%	1	0	0	0	0
threaded_rod	2	8.7%	0	2	0	0	0
channel	1	4.3%	0	1	0	0	0
ceiling_anchor	2	8.7%	0	2	0	0	2

Manufacturer by element type

manufacturer	pipe	duct_round	duct_rectangular	duct_flex	radiator	threaded_rod	channel	ceiling_anchor	products
Hilti	3	0	0	0	0	2	1	0	6
Kermi	0	0	0	0	1	0	0	0	1
Müpro	3	0	0	0	0	0	0	0	3
Sikla	0	1	2	0	0	0	0	0	3
Walraven	2	0	0	0	0	0	0	0	2

manufacturer	pipe	duct_round	duct_rectangular	duct_flex	radiator	threaded_rod	channel	ceiling_anchor	products
fischer	6	0	0	0	0	0	0	2	8
all	14	1	2	0	1	2	1	2	23

Products by role

role	products	share	cumulative share
primary	18	78.3%	78.3%
companion	5	21.7%	100.0%

Field completeness

Filled means the JSON value is not null and, for a list, not empty. max_support_spacing is null on every product: published spacing lives in guidance.

field	filled	empty	filled share
product_line	23	0	100.0%
article_number	23	0	100.0%
outer_diameter_mm	14	9	60.9%
nominal_diameter_mm	1	22	4.3%
connection_threads	22	1	95.7%
load_rating	13	10	56.5%
max_support_spacing	0	23	0.0%
anchor	3	20	13.0%
assumptions	3	20	13.0%
notes	23	0	100.0%
sources	23	0	100.0%

Article numbers: 23 stored, 23 distinct. Product ids: 23 stored, 23 distinct.

Connection threads

A clamp with M8/M10 is counted under both threads. The share is the share of catalogue products that include the thread, so the column can sum to more than 100%.

thread	products	share of catalogue
M10	19	82.6%
M8	15	65.2%
M12	4	17.4%

Products with an empty connection_threads list: 1 of 23.

thread combination	role	products	share of catalogue
(none)	primary	1	4.3%
M10	companion	2	8.7%

thread combination	role	products	share of catalogue
M10	primary	1	4.3%
M10+M12	primary	3	13.0%
M12	companion	1	4.3%
M8	companion	2	8.7%
M8+M10	primary	13	56.5%

Pipe-clamp windows

id	manufacturer	article	min mm	max mm	span mm	threads	load kN
fischer-frs-l-539443	fischer	539443	8	11	3	M8, M10	0.7
fischer-frs-plus-079440	fischer	079440	12	15	3	M8, M10	0.8
muepro-optimal-119866	Müpro	119866	13	15	2	M8, M10	
fischer-frs-plus-079441	fischer	079441	15	19	4	M8, M10	0.8
hilti-mp-mi-20843	Hilti	20843	15	19	4	M10, M12	1.8
muepro-optimal-119862	Müpro	119862	19	24	5	M8, M10	
fischer-frs-plus-079442	fischer	079442	20	24	4	M8, M10	0.8
hilti-mp-mi-20845	Hilti	20845	20	25	5	M10, M12	1.8
fischer-frs-plus-079445	fischer	079445	40	45	5	M8, M10	1
hilti-mp-mi-20851	Hilti	20851	40	45	5	M10, M12	1.8
muepro-optimal-119858	Müpro	119858	41	44	3	M8, M10	
walraven-bismat-2000-3403105	Walraven	3403105	100	105	5	M8, M10	
walraven-bismat-ksb2-3396105	Walraven	3396105	100	105	5	M8, M10	
fischer-frs-a2-064697	fischer	064697	102	116	14	M10	

Span is max minus min of the published window. Load kN converts N by dividing by 1000 and leaves a cell empty when the source published no load. The definitions still differ by product and are not one allowable-load population.

Descriptive statistics of pipe-clamp windows

series	n	sum	min	max	range	mean	median	sample stdev	q1	q3
window min, mm	14	545	8	102	94	38.929	20	35.145	15	40.75
window max, mm	14	612	11	116	105	43.714	24.5	37.012	19	45
window span, mm	14	67	2	14	12	4.786	4.5	2.833	3.25	5
published load of these clamps, kN	8	9.5	0.7	1.8	1.1	1.188	0.9	0.514	0.8	1.8

Overlapping pipe-clamp windows

Two windows overlap when their intervals meet within 0.02 mm, the same tolerance the mapper uses. A zero-width overlap is a shared boundary, so an outside diameter equal to that boundary matches both clamps.

article A	article B	overlap min mm	overlap max mm	overlap width mm
119866	079440	13	15	2
079440	079441	15	15	0
079440	20843	15	15	0
079441	20843	15	19	4
119866	079441	15	15	0
119866	20843	15	15	0
119862	079441	19	19	0
119862	20843	19	19	0
079442	20845	20	24	4
119862	079442	20	24	4
119862	20845	20	24	4
079445	20851	40	45	5
119858	079445	41	44	3
119858	20851	41	44	3
3396105	3403105	100	105	5
064697	3396105	102	105	3
064697	3403105	102	105	3

Overlapping or boundary-touching pairs: 17.

Merged clamp coverage and holes

covered from mm	covered to mm	width mm
8	11	3
12	25	13
40	45	5
100	116	16

hole after mm	hole before mm	width mm
11	12	1
25	40	15
45	100	55

Depth is how many primary pipe clamps the mapper returns for each integer millimetre from the lowest published minimum to the highest published maximum.

Descriptive statistics of integer clamp depth

series	n	sum	min	max	range	mean	median	sample stdev	q1	q3
clamps covering each integer mm	109	81	0	4	4	0.743	0	1.117	0	1

clamps at that mm	integer mm count	share of the scanned mm	cumulative share
0	68	62.4%	62.4%
1	17	15.6%	78.0%
3	14	12.8%	90.8%
2	9	8.3%	99.1%
4	1	0.9%	100.0%

uncovered integers from mm	uncovered integers to mm	count of integers
26	39	14
46	99	54

Clamp depth at each integer millimetre

mm	matching clamps	product ids
8	1	fischer-frs-l-539443
9	1	fischer-frs-l-539443
10	1	fischer-frs-l-539443
11	1	fischer-frs-l-539443
12	1	fischer-frs-plus-079440
13	2	muepro-optimal-119866, fischer-frs-plus-079440
14	2	muepro-optimal-119866, fischer-frs-plus-079440
15	4	muepro-optimal-119866, fischer-frs-plus-079440, fischer-frs-plus-079441, hilti-mp-mi-20843
16	2	fischer-frs-plus-079441, hilti-mp-mi-20843
17	2	fischer-frs-plus-079441, hilti-mp-mi-20843
18	2	fischer-frs-plus-079441, hilti-mp-mi-20843
19	3	muepro-optimal-119862, fischer-frs-plus-079441, hilti-mp-mi-20843
20	3	muepro-optimal-119862, fischer-frs-plus-079442, hilti-mp-mi-20845
21	3	muepro-optimal-119862, fischer-frs-plus-079442, hilti-mp-mi-20845
22	3	muepro-optimal-119862, fischer-frs-plus-079442, hilti-mp-mi-20845
23	3	muepro-optimal-119862, fischer-frs-plus-079442, hilti-mp-mi-20845
24	3	muepro-optimal-119862, fischer-frs-plus-079442, hilti-mp-mi-20845
25	1	hilti-mp-mi-20845
26	0	
27	0	
28	0	
29	0	
30	0	
31	0	
32	0	

mm	matching clamps	product ids
33	0	
34	0	
35	0	
36	0	
37	0	
38	0	
39	0	
40	2	fischer-frs-plus-079445, hilti-mp-mi-20851
41	3	muepro-optimal-119858, fischer-frs-plus-079445, hilti-mp-mi-20851
42	3	muepro-optimal-119858, fischer-frs-plus-079445, hilti-mp-mi-20851
43	3	muepro-optimal-119858, fischer-frs-plus-079445, hilti-mp-mi-20851
44	3	muepro-optimal-119858, fischer-frs-plus-079445, hilti-mp-mi-20851
45	2	fischer-frs-plus-079445, hilti-mp-mi-20851
46	0	
47	0	
48	0	
49	0	
50	0	
51	0	
52	0	
53	0	
54	0	
55	0	
56	0	
57	0	
58	0	
59	0	
60	0	
61	0	
62	0	
63	0	
64	0	
65	0	
66	0	
67	0	
68	0	
69	0	
70	0	

mm	matching clamps	product ids
71	0	
72	0	
73	0	
74	0	
75	0	
76	0	
77	0	
78	0	
79	0	
80	0	
81	0	
82	0	
83	0	
84	0	
85	0	
86	0	
87	0	
88	0	
89	0	
90	0	
91	0	
92	0	
93	0	
94	0	
95	0	
96	0	
97	0	
98	0	
99	0	
100	2	walraven-bismat-ksb2-3396105, walraven-bismat-2000-3403105
101	2	walraven-bismat-ksb2-3396105, walraven-bismat-2000-3403105
102	3	fischer-frs-a2-064697, walraven-bismat-ksb2-3396105, walraven-bismat-2000-3403105
103	3	fischer-frs-a2-064697, walraven-bismat-ksb2-3396105, walraven-bismat-2000-3403105
104	3	fischer-frs-a2-064697, walraven-bismat-ksb2-3396105, walraven-bismat-2000-3403105
105	3	fischer-frs-a2-064697, walraven-bismat-ksb2-3396105, walraven-bismat-2000-3403105
106	1	fischer-frs-a2-064697
107	1	fischer-frs-a2-064697
108	1	fischer-frs-a2-064697

mm	matching clamps	product ids
109	1	fischer-frs-a2-064697
110	1	fischer-frs-a2-064697
111	1	fischer-frs-a2-064697
112	1	fischer-frs-a2-064697
113	1	fischer-frs-a2-064697
114	1	fischer-frs-a2-064697
115	1	fischer-frs-a2-064697
116	1	fischer-frs-a2-064697

Published loads

kN is the stored value when the unit is already kN, and the stored value divided by 1000 when the unit is N. Empty load cells are excluded from the descriptive rows. Kinds are kept apart because a recommended static load, a Nutzlast and an anchor Fzul are not interchangeable.

id	manufacturer	element type	value	unit	kN	kind
fischer-frs-l-539443	fischer	pipe	0.7	kN	0.7	max. recommended static load, centric tension
fischer-frs-plus-079440	fischer	pipe	0.8	kN	0.8	max. recommended static load, centric tension
fischer-frs-plus-079441	fischer	pipe	0.8	kN	0.8	max. recommended static load, centric tension
fischer-frs-plus-079442	fischer	pipe	0.8	kN	0.8	max. recommended static load, centric tension
fischer-frs-plus-079445	fischer	pipe	1	kN	1	max. recommended static load, centric tension
hilti-mp-mi-20843	Hilti	pipe	1800	N	1.8	maximum load F, ambient, for the 15–64 mm band of MP-MI series G
hilti-mp-mi-20845	Hilti	pipe	1800	N	1.8	maximum load F, ambient, for the 15–64 mm band of MP-MI series G
hilti-mp-mi-20851	Hilti	pipe	1800	N	1.8	maximum load F, ambient, for the 15–64 mm band of MP-MI series G
sikla-plu-2g-199893	Sikla	duct_round	0.8	kN	0.8	zulässige Nutzlast for types 71–200
sikla-luw-sl-189893	Sikla	duct_rectangular	0.8	kN	0.8	Nennlast
sikla-luw-sz-189902	Sikla	duct_rectangular	0.8	kN	0.8	Nennlast
fischer-ea-ii-m8-532231	fischer	ceiling_anchor	1.4	kN	1.4	permissible load Fzul of one anchor for redundant non-structural use in concrete C20/25 to C50/60
fischer-ea-ii-m10-532232	fischer	ceiling_anchor	1.9	kN	1.9	permissible load Fzul of one anchor for redundant non-structural use in concrete C20/25 to C50/60

series	n	sum	min	max	range	mean	median	sample stdev	q1	q3
all published loads converted to kN	13	15.2	0.7	1.9	1.2	1.169	0.8	0.487	0.8	1.8
Nennlast	2	1.6	0.8	0.8	0	0.8	0.8	0		
max. recommended static load, centric tension	5	4.1	0.7	1	0.3	0.82	0.8	0.11	0.8	0.8
maximum load F, ambient, for the 15–64 mm band of MP-MI series G	3	5.4	1.8	1.8	0	1.8	1.8	0		
permissible load Fzul of one anchor for redundant non-structural use in concrete C20/25 to C50/60	2	3.3	1.4	1.9	0.5	1.65	1.65	0.354		
zulässige Nutzlast for types 71–200	1	0.8	0.8	0.8	0	0.8	0.8			

id	manufacturer	element type	article
muepro-optimal-119866	Müpro	pipe	119866
muepro-optimal-119862	Müpro	pipe	119862
muepro-optimal-119858	Müpro	pipe	119858
fischer-frs-a2-064697	fischer	pipe	064697
walraven-bismat-ksb2-3396105	Walraven	pipe	3396105
walraven-bismat-2000-3403105	Walraven	pipe	3403105
hilti-mq-41-3-369596	Hilti	channel	369596
hilti-rod-am8-339793	Hilti	threaded_rod	339793
hilti-rod-am10-339795	Hilti	threaded_rod	339795
kermi-console-zb02870003	Kermi	radiator	ZB02870003

Products with no published load in the database: 10 of 23.

Anchors stored on products

id	manufacturer	anchor type	thread	substrate	included fixings	load kN
fischer-ea-ii-m8-532231	fischer	hammer-set internally threaded anchor	M8	concrete		1.4
fischer-ea-ii-m10-532232	fischer	hammer-set internally threaded anchor	M10	concrete		1.9
kermi-console-zb02870003	Kermi	wall plug and screw, included with the console set		wall	2 screws 8×60 mm, 2 plugs 10×60 mm, 2 washers	

substrate	products	share of anchored products	cumulative share
concrete	2	66.7%	66.7%
wall	1	33.3%	100.0%

Guidance records

id	applies to	rows	rows flagged table_row	sources	note characters
copper-tube-spacing-din1988-2-as-cited	pipe	12		2	610
lindab-duct-suspension-spacing	duct_round, duct_rectangular	6	4	1	292
en-12236-duct-support-strength	duct_round, duct_rectangular, duct_flex	0		2	557

Guidance rows: copper-tube-spacing-din1988-2-as-cited

Both catalogues print this as Richtwerte for copper and stainless tube and attribute it to DIN 1988 Teil 2. The standard text was not read for this database. The column copied here is straight length (Stangenrohr). The same IBP table gives shorter distances for copper coil (Ringrohr), for example 0.7–0.9 m at 15 mm, and those are not applied. No vertical-stack column is in this table. A row is offered for a modeled pipe only when the outside diameter is within 0.5 mm of a listed value, and only with the explicit assumption that the unnamed IFC material is the copper or stainless tube the table is about.

outer_diameter_mm	horizontal_m	form
12	1.25	stangenrohr
15	1.25	stangenrohr

outer_diameter_mm	horizontal_m	form
18	1.5	stangenrohr
22	2	stangenrohr
28	2.25	stangenrohr
35	2.75	stangenrohr
42	3	stangenrohr
54	3.5	stangenrohr
64	4	stangenrohr
76.1	4.25	stangenrohr
88.9	4.75	stangenrohr
108	5	stangenrohr

series	n	sum	min	max	range	mean	median	sample stdev	q1	q3
horizontal spacing, m	12	35.5	1.25	5	3.75	2.958	2.875	1.343	1.875	4.062

series	n	sum	min	max	range	mean	median	sample stdev	q1	q3
table outside diameter, mm	12	563	12	108	96	46.917	38.5	31.488	21	67.025

Guidance rows: lindab-duct-suspension-spacing

Swedish Lindab suspension guide, widely used for European spiral and rectangular duct. Vertical runs in the same section are limited to 3 m for both shapes. The size-table rows above are the uninsulated, no-fire-class case. Insulated and fire-rated tables in the same document are not copied.

shape	diameter_mm	horizontal_m	table_row	conditions	width_mm	height_mm
round	150	3	yes	uninsulated circular duct, no fire class; recommended spacing 3.0 m across the listed hanger types		
rectangular		2.4	yes	uninsulated rectangular duct, no fire class	200	150
rectangular		2.4	yes	uninsulated rectangular duct, no fire class	300	150
rectangular		2.4	yes	uninsulated rectangular duct, no fire class	300	300
rectangular		2.4	no	stated maximum hanger spacing for horizontal rectangular duct in that section, because more than one duct joint between hangers is considered unsuitable. 150×150 mm is not a row in the size table, which starts at width 200 mm.		
round		3	no	stated maximum hanger spacing for horizontal circular duct in that section		

series	n	sum	min	max	range	mean	median	sample stdev	q1	q3
horizontal spacing of table_row true, m	4	10.2	2.4	3	0.6	2.55	2.4	0.3	2.4	2.55

Guidance rows: en-12236-duct-support-strength

DIN EN 12236:2002 (EN 12236:2002) specifies strength of supports for sheet-metal ductwork of any shape. It does not publish a spacing table. From the preview of the EN text: the fixing into the structure needs a safety factor of at least 3 on the calculated load; vertical members and the joint to the horizontal member need a safety factor of at least 1.5 against yield; horizontal-member deflection must not exceed 0.4 % of the distance between the vertical fixings. Fire loading is outside the

standard's scope. No spacing number is derived from it here.

This record has no numeric rows.

Inventory by model

model	file	schema	elements	share of elements	groups	recorded length m	elements matched	elements with assumption	elements unmatched	element matched share	element unmatched share	groups matched	groups with assumption	groups unmatched
piping	zenodo-duplex-piping.ifc	IFC2X3	463	39.8%	41	108.565	449	0	14	97.0%	3.0%	37	0	4
ventilation	zenodo-duplex-ventilation.ifc	IFC2X3	374	32.2%	40	118.376	20	274	80	5.3%	21.4%	3	29	8
heating_and_cooling	HeatingAndCooling.ifc	IFC2X3	326	28.0%	17	195.267	308	16	2	94.5%	0.6%	15	1	1

Elements by kind

kind	elements	share of elements	cumulative share
pipe_fitting	386	33.2%	33.2%
pipe_segment	357	30.7%	63.9%
duct_fitting	156	13.4%	77.3%
duct_segment	126	10.8%	88.1%
air_terminal	40	3.4%	91.6%
duct_flex	36	3.1%	94.7%
radiator	16	1.4%	96.0%
pipe_accessory	14	1.2%	97.2%
duct_accessory	14	1.2%	98.5%
sanitary_fixture	12	1.0%	99.5%
water_heater	2	0.2%	99.7%
air_handling_unit	2	0.2%	99.8%
stove	2	0.2%	100.0%

Size groups by kind

kind	groups	share of groups	cumulative share
pipe_fitting	40	40.8%	40.8%
duct_fitting	25	25.5%	66.3%
pipe_segment	9	9.2%	75.5%
duct_segment	5	5.1%	80.6%
air_terminal	4	4.1%	84.7%
pipe_accessory	3	3.1%	87.8%

kind	groups	share of groups	cumulative share
sanitary_fixture	3	3.1%	90.8%
duct_accessory	3	3.1%	93.9%
duct_flex	2	2.0%	95.9%
water_heater	1	1.0%	96.9%
air_handling_unit	1	1.0%	98.0%
radiator	1	1.0%	99.0%
stove	1	1.0%	100.0%

kind	piping	ventilation	heating_and_cooling	elements
air_handling_unit	0	2	0	2
air_terminal	0	40	0	40
duct_accessory	0	14	0	14
duct_fitting	0	156	0	156
duct_flex	0	36	0	36
duct_segment	0	126	0	126
pipe_accessory	10	0	4	14
pipe_fitting	233	0	153	386
pipe_segment	206	0	151	357
radiator	0	0	16	16
sanitary_fixture	12	0	0	12
stove	0	0	2	2
water_heater	2	0	0	2

Elements by IFC class

IFC class	elements	share of elements	cumulative share
IfcFlowFitting	526	45.2%	45.2%
IfcFlowSegment	519	44.6%	89.9%
IfcFlowTerminal	84	7.2%	97.1%
IfcFlowController	28	2.4%	99.5%
IfcBuildingElementProxy	4	0.3%	99.8%
IfcEnergyConversionDevice	2	0.2%	100.0%

Elements by size method

size method	elements	share of elements	cumulative share
connected_segment_profile	400	34.4%	34.4%
circle_profile	367	31.6%	66.0%

size method	elements	share of elements	cumulative share
connected_duct_segment	168	14.4%	80.4%
port_axis_against_extrusion	116	10.0%	90.4%
world_bbox	74	6.4%	96.7%
faceted_bbox_minor_axis	20	1.7%	98.5%
unresolved	18	1.5%	100.0%

Elements by family

family	elements	share of elements	cumulative share
(none)	399	34.3%	34.3%
elbow	229	19.7%	54.0%
transition	209	18.0%	72.0%
rectangular	116	10.0%	81.9%
tee	88	7.6%	89.5%
round_flex	36	3.1%	92.6%
supply	20	1.7%	94.3%
return	18	1.5%	95.9%
panel	16	1.4%	97.2%
volume_damper	14	1.2%	98.5%
round	10	0.9%	99.3%
cross	4	0.3%	99.7%
cap	2	0.2%	99.8%
exhaust	2	0.2%	100.0%

Elements by cross-section shape

shape	elements	share of elements	cumulative share
(none)	644	55.4%	55.4%
round	403	34.7%	90.0%
rectangular	116	10.0%	100.0%

Missing material, insulation and predefined type

Counts are size groups. The inventory found no material name and no insulation on these models.

model	groups	groups with no material	groups with no insulation	groups with no predefined type	elements with no material
piping	41	41	41	41	463
ventilation	40	40	40	40	374
heating_and_cooling	17	17	17	17	326

Size groups whose NominalDiameter property disagrees with the connected profile: 23 groups, 243 elements. Size groups with more than one outside diameter (reducers and similar): 17 groups, 132 elements.

System assignments

The count is the sum of system memberships stored on each size group. It can exceed the element count.

model	system object type	assignments	share of that model's assignments	elements in the model	assignments minus elements
piping	Domestic Cold Water	199	41.0%	463	22
piping	Sanitary	170	35.1%	463	22
piping	Domestic Hot Water	116	23.9%	463	22
ventilation	Powietrze nawiewane	188	49.3%	374	7
ventilation	Powietrze recyrkulacyjne	186	48.8%	374	7
ventilation	Powietrze usuwane	6	1.6%	374	7
ventilation	Kanalizacja	1	0.3%	374	7
heating_and_cooling	Hydronic Return	178	51.7%	326	18
heating_and_cooling	Hydronic Supply	166	48.3%	326	18

Pipe segments by outside diameter

model	OD mm	segments	share of segments	length m	share of pipe length	cumulative length share	mean length per segment m	match status	clamps	articles
heating_and_cooling	9.52	81	22.7%	130.103	42.8%	42.8%	1.606	matched	1	539443
heating_and_cooling	12.7	50	14.0%	60.046	19.8%	62.6%	1.201	matched	1	079440
piping	22.225	76	21.3%	43.338	14.3%	76.8%	0.57	matched	3	119862, 079442, 20845
piping	41.275	38	10.6%	23.516	7.7%	84.6%	0.619	matched	3	119858, 079445, 20851
piping	15.875	54	15.1%	18.446	6.1%	90.7%	0.342	matched	2	079441, 20843
piping	104.775	24	6.7%	16.888	5.6%	96.2%	0.704	matched	3	064697, 3396105, 3403105
piping	12.7	6	1.7%	6.121	2.0%	98.2%	1.02	matched	1	079440
heating_and_cooling	19.05	20	5.6%	5.118	1.7%	99.9%	0.256	matched	1	119862
piping	15	8	2.2%	0.256	0.1%	100.0%	0.032	matched	4	119866, 079440, 079441, 20843

Pipe-segment diameter statistics

Min, max, mean, median and sample stdev treat one segment as one observation, so a common small diameter pulls the mean down. The length-weighted columns weight each size by its recorded centreline length.

model	segments	length m	mean length per segment m	min OD mm	max OD mm	count-weighted mean OD mm	count-weighted median OD mm	count-weighted sample stdev mm	length-weighted mean OD mm	length-weighted median OD mm
piping	206	108.565	0.527	12.7	104.775	33.134	22.225	27.582	37.56	22.225
heating_and_cooling	151	195.267	1.293	9.52	19.05	11.835	9.52	3.176	10.748	9.52
all pipe segments	357	303.832	0.851	9.52	104.775	24.125	15.875	23.524	20.328	12.7

series	n	sum	min	max	range	mean	median	sample stdev	q1	q3
distinct pipe-segment OD values, one row per size group, mm	9	253.12	9.52	104.775	95.255	28.124	15.875	30.228	12.7	22.225

Duct segments by cross-section

model	shape	size	width mm	height mm	OD mm	section area mm2	segments	length m	share of duct length	cumulative length share	mean length per segment m	match status	primary products
ventilation	rectangular	200.0 × 150.0 mm	200	150		30000	46	43.655	36.9%	36.9%	0.949	matched_with_assumption	2
ventilation	rectangular	150.0 × 150.0 mm	150	150		22500	30	31.835	26.9%	63.8%	1.061	matched_with_assumption	2
ventilation	rectangular	300.0 × 300.0 mm	300	300		90000	22	21.939	18.5%	82.3%	0.997	matched_with_assumption	2
ventilation	rectangular	300.0 × 150.0 mm	300	150		45000	18	11.051	9.3%	91.6%	0.614	matched_with_assumption	2
ventilation	round	OD 150.0 mm			150		10	9.896	8.4%	100.0%	0.99	matched	1

series	n	sum	min	max	range	mean	median	sample stdev	q1	q3
rectangular duct width, one size group, mm	4	950	150	300	150	237.5	250	75	187.5	300
rectangular duct height, one size group, mm	4	750	150	300	150	187.5	150	75	150	187.5
rectangular section area, one size group, mm2	4	187500	22500	90000	67500	46875	37500	30233.467	28125	56250
duct segment length, one size group, m	5	118.376	9.896	43.655	33.759	23.675	21.939	14.301	11.051	31.835

Length-weighted mean rectangular width is 215.738 mm; height 180.336 mm; section area 41461.491 mm2.

Devices and other bounding boxes

model	kind	IFC class	size	elements	share of all elements	bbox axis 1 mm	bbox axis 2 mm	bbox axis 3 mm	status	gap
piping	sanitary_fixture	IfcFlowTerminal	bbox 228.2 × 510 × 635 mm	4	0.3%	228.2	510	635	unmatched	No sanitary carrier or fixture bracket is in the database.
piping	sanitary_fixture	IfcFlowTerminal	bbox 361.9 × 584.3 × 722.5 mm	4	0.3%	361.9	584.3	722.5	unmatched	No sanitary carrier or fixture bracket is in the database.
piping	sanitary_fixture	IfcFlowTerminal	bbox 457 × 457.2 × 485.8 mm	4	0.3%	457	457.2	485.8	unmatched	No sanitary carrier or fixture bracket is in the database.

model	kind	IFC class	size	elements	share of all elements	bbox axis 1 mm	bbox axis 2 mm	bbox axis 3 mm	status	gap
piping	water_heater	IfcBuildingElementProxy	bbox 350 × 500 × 1000 mm	2	0.2%	350	500	1000	unmatched	No water-heater support is in the database. The proxy is a floor-standing appliance, not a suspended pipe.
ventilation	air_handling_unit	IfcBuildingElementProxy	bbox 475 × 1100 × 1400 mm	2	0.2%	475	1100	1400	unmatched	No air-handling-unit frame is in the database.
ventilation	air_terminal	IfcFlowTerminal	bbox 101.6 × 200 × 200 mm	18	1.5%	101.6	200	200	unmatched	No diffuser or grille fixing is in the database.
ventilation	air_terminal	IfcFlowTerminal	bbox 101.6 × 200 × 200 mm	18	1.5%	101.6	200	200	unmatched	No diffuser or grille fixing is in the database.
ventilation	air_terminal	IfcFlowTerminal	bbox 40 × 300 × 300 mm	2	0.2%	40	300	300	unmatched	No diffuser or grille fixing is in the database.
ventilation	air_terminal	IfcFlowTerminal	bbox 40 × 300 × 300 mm	2	0.2%	40	300	300	unmatched	No diffuser or grille fixing is in the database.
heating_and_cooling	radiator	IfcFlowTerminal	bbox 100 × 350 × 1000 mm	16	1.4%	100	350	1000	matched_with_assumption	
heating_and_cooling	stove	IfcEnergyConversionDevice	bbox 350 × 500 × 1000 mm	2	0.2%	350	500	1000	unmatched	No stove or boiler support is in the database. The element is floor-standing equipment.

Coverage by kind

kind	groups	elements	matched elements	matched share	assumption elements	assumption share	unmatched elements	unmatched share
air_handling_unit	1	2	0	0.0%	0	0.0%	2	100.0%
air_terminal	4	40	0	0.0%	0	0.0%	40	100.0%
duct_accessory	3	14	0	0.0%	14	100.0%	0	0.0%
duct_fitting	25	156	10	6.4%	144	92.3%	2	1.3%
duct_flex	2	36	0	0.0%	0	0.0%	36	100.0%
duct_segment	5	126	10	7.9%	116	92.1%	0	0.0%
pipe_accessory	3	14	14	100.0%	0	0.0%	0	0.0%
pipe_fitting	40	386	386	100.0%	0	0.0%	0	0.0%
pipe_segment	9	357	357	100.0%	0	0.0%	0	0.0%
radiator	1	16	0	0.0%	16	100.0%	0	0.0%
sanitary_fixture	3	12	0	0.0%	0	0.0%	12	100.0%
stove	1	2	0	0.0%	0	0.0%	2	100.0%
water_heater	1	2	0	0.0%	0	0.0%	2	100.0%

Unmatched size groups

model	kind	size	elements	share of that model	share of all elements	gap
piping	sanitary_fixture	bbox 228.2 × 510 × 635 mm	4	0.9%	0.3%	No sanitary carrier or fixture bracket is in the database.
piping	sanitary_fixture	bbox 361.9 × 584.3 × 722.5 mm	4	0.9%	0.3%	No sanitary carrier or fixture bracket is in the database.
piping	sanitary_fixture	bbox 457 × 457.2 × 485.8 mm	4	0.9%	0.3%	No sanitary carrier or fixture bracket is in the database.
piping	water_heater	bbox 350 × 500 × 1000 mm	2	0.4%	0.2%	No water-heater support is in the database. The proxy is a floor-standing appliance, not a suspended pipe.
ventilation	air_handling_unit	bbox 475 × 1100 × 1400 mm	2	0.5%	0.2%	No air-handling-unit frame is in the database.
ventilation	air_terminal	bbox 101.6 × 200 × 200 mm	18	4.8%	1.5%	No diffuser or grille fixing is in the database.
ventilation	air_terminal	bbox 101.6 × 200 × 200 mm	18	4.8%	1.5%	No diffuser or grille fixing is in the database.
ventilation	air_terminal	bbox 40 × 300 × 300 mm	2	0.5%	0.2%	No diffuser or grille fixing is in the database.
ventilation	air_terminal	bbox 40 × 300 × 300 mm	2	0.5%	0.2%	No diffuser or grille fixing is in the database.
ventilation	duct_fitting	connected duct unresolved	2	0.5%	0.2%	No connected duct size was recovered, so no duct hanger can be selected.
ventilation	duct_flex	mesh minor axis 151.6 mm	20	5.3%	1.7%	No product has a clamping range covering flex-duct mesh minor axis 151.6 mm. Sikla PLU type 150 is a spiral-seam clamp for nominal 150 mm and is not applied.
ventilation	duct_flex	diameter unresolved	16	4.3%	1.4%	Flex-duct diameter could not be recovered from the mesh, and no flex-duct hanger is in the database.
heating_and_cooling	stove	bbox 350 × 500 × 1000 mm	2	0.6%	0.2%	No stove or boiler support is in the database. The element is floor-standing equipment.

Unmatched groups: 13. Unmatched elements: 96 (8.3% of 1163).

Primary-product utilisation

Element hits sum the size groups for which the product is a primary match. A reducer that matches two clamps, and a diameter that matches three clamps, are counted once for each product. The shares can therefore sum to more than 100%.

id	manufacturer	article	element type	groups	element hits	share of all elements
sikla-luw-sl-189893	Sikla	189893	duct_rectangular	29	274	23.6%
sikla-luw-sz-189902	Sikla	189902	duct_rectangular	29	274	23.6%
muepro-optimal-119862	Müpro	119862	pipe	18	252	21.7%
fischer-frs-plus-079442	fischer	079442	pipe	12	200	17.2%
hilti-mp-mi-20845	Hilti	20845	pipe	12	200	17.2%
fischer-frs-l-539443	fischer	539443	pipe	7	184	15.8%
fischer-frs-plus-079440	fischer	079440	pipe	16	174	15.0%
fischer-frs-plus-079441	fischer	079441	pipe	12	147	12.6%
hilti-mp-mi-20843	Hilti	20843	pipe	12	147	12.6%
fischer-frs-plus-079445	fischer	079445	pipe	10	84	7.2%
hilti-mp-mi-20851	Hilti	20851	pipe	10	84	7.2%
muepro-optimal-119858	Müpro	119858	pipe	10	84	7.2%
fischer-frs-a2-064697	fischer	064697	pipe	8	60	5.2%
walraven-bismat-2000-3403105	Walraven	3403105	pipe	8	60	5.2%

id	manufacturer	article	element type	groups	element hits	share of all elements
walraven-bismat-ksb2-3396105	Walraven	3396105	pipe	8	60	5.2%
sikla-plu-2g-199893	Sikla	199893	duct_round	7	34	2.9%
muepro-optimal-119866	Müpro	119866	pipe	3	20	1.7%
kermi-console-zb02870003	Kermi	ZB02870003	radiator	1	16	1.4%

Primary products with no hit in this inventory: none. Primary products with at least one hit: 18 of 18.

series	n	sum	min	max	range	mean	median	sample stdev	q1	q3
element hits per primary product	18	2354	16	274	258	130.778	115.5	86.41	60	196

Companion products under two substrates

Companions are rods, channel and anchors returned with a primary match. substrate none is the duplex default. substrate concrete is the German commercial ceiling the anchor articles were recorded for. Each element is counted once per companion: a rod is attached when the matched clamps use that thread, and it is not multiplied by the number of clamps.

id	element type	threads	groups, substrate none	elements, substrate none	groups, substrate concrete	elements, substrate concrete
hilti-mq-41-3-369596	channel	M12	29	274	29	274
hilti-rod-am8-339793	threaded_rod	M8	84	1051	84	1051
hilti-rod-am10-339795	threaded_rod	M10	84	1051	84	1051
fischer-ea-ii-m8-532231	ceiling_anchor	M8	0	0	84	1051
fischer-ea-ii-m10-532232	ceiling_anchor	M10	0	0	84	1051

Size-group census

model	kind	family	IFC class	size	elements	length m	size method	shape	status	primary products	assumption notes
piping	pipe_accessory	(none)	IfcFlowController	OD 22.225 mm	8		connected_segment_profile	(none)	matched	3	0
piping	pipe_accessory	(none)	IfcFlowController	OD 15.875 mm	2		connected_segment_profile	(none)	matched	2	0
piping	pipe_fitting	elbow	IfcFlowFitting	OD 22.225 mm	38		connected_segment_profile	(none)	matched	3	1
piping	pipe_fitting	elbow	IfcFlowFitting	OD 15.875 mm	34		connected_segment_profile	(none)	matched	2	1
piping	pipe_fitting	transition	IfcFlowFitting	OD 22.225 mm	22		connected_segment_profile	(none)	matched	3	0
piping	pipe_fitting	elbow	IfcFlowFitting	OD 41.275 mm	20		connected_segment_profile	(none)	matched	3	1
piping	pipe_fitting	transition	IfcFlowFitting	OD 15.875 / 22.225 mm	18		connected_segment_profile	(none)	matched	5	0
piping	pipe_fitting	elbow	IfcFlowFitting	OD 104.775 mm	12		connected_segment_profile	(none)	matched	3	1
piping	pipe_fitting	tee	IfcFlowFitting	OD 15.875 / 22.225 mm	10		connected_segment_profile	(none)	matched	5	1
piping	pipe_fitting	transition	IfcFlowFitting	OD 22.225 / 104.775 mm	10		connected_segment_profile	(none)	matched	6	0
piping	pipe_fitting	tee	IfcFlowFitting	OD 22.225 mm	10		connected_segment_profile	(none)	matched	3	1

model	kind	family	IFC class	size	elements	length m	size method	shape	status	primary products	assumption notes
piping	pipe_fitting	transition	IfcFlowFitting	OD 15 / 15.875 mm	8		connected_segment_profile	(none)	matched	4	0
piping	pipe_fitting	tee	IfcFlowFitting	OD 41.275 mm	6		connected_segment_profile	(none)	matched	3	1
piping	pipe_fitting	transition	IfcFlowFitting	OD 15.875 mm	5		connected_segment_profile	(none)	matched	2	0
piping	pipe_fitting	elbow	IfcFlowFitting	OD 12.7 mm	4		connected_segment_profile	(none)	matched	1	0
piping	pipe_fitting	transition	IfcFlowFitting	OD 15 / 41.275 mm	4		connected_segment_profile	(none)	matched	7	0
piping	pipe_fitting	tee	IfcFlowFitting	OD 41.275 / 104.775 mm	4		connected_segment_profile	(none)	matched	6	1
piping	pipe_fitting	transition	IfcFlowFitting	OD 41.275 / 104.775 mm	4		connected_segment_profile	(none)	matched	6	0
piping	pipe_fitting	tee	IfcFlowFitting	OD 104.775 mm	2		connected_segment_profile	(none)	matched	3	1
piping	pipe_fitting	(none)	IfcFlowFitting	OD 104.775 mm	2		connected_segment_profile	(none)	matched	3	1
piping	pipe_fitting	tee	IfcFlowFitting	OD 12.7 / 41.275 mm	2		connected_segment_profile	(none)	matched	4	0
piping	pipe_fitting	transition	IfcFlowFitting	OD 12.7 / 41.275 mm	2		connected_segment_profile	(none)	matched	4	0
piping	pipe_fitting	(none)	IfcFlowFitting	OD 12.7 mm	2		connected_segment_profile	(none)	matched	1	1
piping	pipe_fitting	elbow	IfcFlowFitting	OD 15.875 / 22.225 mm	2		connected_segment_profile	(none)	matched	5	1
piping	pipe_fitting	tee	IfcFlowFitting	OD 22.225 / 104.775 mm	2		connected_segment_profile	(none)	matched	6	1
piping	pipe_fitting	(none)	IfcFlowFitting	OD 22.225 mm	2		connected_segment_profile	(none)	matched	3	1
piping	pipe_fitting	(none)	IfcFlowFitting	OD 22.225 mm	2		connected_segment_profile	(none)	matched	3	1
piping	pipe_fitting	transition	IfcFlowFitting	OD 41.275 mm	2		connected_segment_profile	(none)	matched	3	0
piping	pipe_fitting	cap	IfcFlowTerminal	OD 41.275 mm	2		connected_segment_profile	(none)	matched	3	0
piping	pipe_fitting	(none)	IfcFlowFitting	OD 15.875 mm	1		connected_segment_profile	(none)	matched	2	1
piping	pipe_fitting	(none)	IfcFlowFitting	OD 15.875 mm	1		connected_segment_profile	(none)	matched	2	1
piping	pipe_segment	(none)	IfcFlowSegment	OD 22.225 mm	76	43.338	circle_profile	round	matched	3	1
piping	pipe_segment	(none)	IfcFlowSegment	OD 15.875 mm	54	18.446	circle_profile	round	matched	2	1
piping	pipe_segment	(none)	IfcFlowSegment	OD 41.275 mm	38	23.516	circle_profile	round	matched	3	1
piping	pipe_segment	(none)	IfcFlowSegment	OD 104.775 mm	24	16.888	circle_profile	round	matched	3	1
piping	pipe_segment	(none)	IfcFlowSegment	OD 15.0 mm	8	0.256	circle_profile	round	matched	4	1
piping	pipe_segment	(none)	IfcFlowSegment	OD 12.7 mm	6	6.121	circle_profile	round	matched	1	1
piping	sanitary_fixture	(none)	IfcFlowTerminal	bbox 228.2 × 510 × 635 mm	4		world_bbox	(none)	unmatched	0	0
piping	sanitary_fixture	(none)	IfcFlowTerminal	bbox 361.9 × 584.3 × 722.5 mm	4		world_bbox	(none)	unmatched	0	0
piping	sanitary_fixture	(none)	IfcFlowTerminal	bbox 457 × 457.2 × 485.8 mm	4		world_bbox	(none)	unmatched	0	0

model	kind	family	IFC class	size	elements	length m	size method	shape	status	primary products	assumption notes
piping	water_heater	(none)	IfcBuildingElementProxy	bbox 350 × 500 × 1000 mm	2		world_bbox	(none)	unmatched	0	0
ventilation	air_handling_unit	(none)	IfcBuildingElementProxy	bbox 475 × 1100 × 1400 mm	2		world_bbox	(none)	unmatched	0	0
ventilation	air_terminal	supply	IfcFlowTerminal	bbox 101.6 × 200 × 200 mm	18		world_bbox	(none)	unmatched	0	0
ventilation	air_terminal	return	IfcFlowTerminal	bbox 101.6 × 200 × 200 mm	18		world_bbox	(none)	unmatched	0	0
ventilation	air_terminal	exhaust	IfcFlowTerminal	bbox 40 × 300 × 300 mm	2		world_bbox	(none)	unmatched	0	0
ventilation	air_terminal	supply	IfcFlowTerminal	bbox 40 × 300 × 300 mm	2		world_bbox	(none)	unmatched	0	0
ventilation	duct_accessory	volume_damper	IfcFlowController	connected 200×150 mm	6		connected_duct_segment	(none)	matched_with_assumption	2	0
ventilation	duct_accessory	volume_damper	IfcFlowController	connected 150×150 mm	4		connected_duct_segment	(none)	matched_with_assumption	2	0
ventilation	duct_accessory	volume_damper	IfcFlowController	connected 300×150 mm	4		connected_duct_segment	(none)	matched_with_assumption	2	0
ventilation	duct_fitting	transition	IfcFlowFitting	connected 200×150 mm	28		connected_duct_segment	(none)	matched_with_assumption	2	0
ventilation	duct_fitting	transition	IfcFlowFitting	connected 150×150 mm	26		connected_duct_segment	(none)	matched_with_assumption	2	0
ventilation	duct_fitting	elbow	IfcFlowFitting	connected 200×150 mm	12		connected_duct_segment	(none)	matched_with_assumption	2	0
ventilation	duct_fitting	elbow	IfcFlowFitting	connected 150×150 mm	10		connected_duct_segment	(none)	matched_with_assumption	2	0
ventilation	duct_fitting	elbow	IfcFlowFitting	connected 300×300 mm	10		connected_duct_segment	(none)	matched_with_assumption	2	0
ventilation	duct_fitting	tee	IfcFlowFitting	connected 200×150 mm	8		connected_duct_segment	(none)	matched_with_assumption	2	0
ventilation	duct_fitting	transition	IfcFlowFitting	connected Ø150 mm	8		connected_duct_segment	(none)	matched	1	0
ventilation	duct_fitting	elbow	IfcFlowFitting	connected 300×150 mm	6		connected_duct_segment	(none)	matched_with_assumption	2	0

model	kind	family	IFC class	size	elements	length m	size method	shape	status	primary products	assumption notes
ventilation	duct_fitting	transition	IfcFlowFitting	connected Ø150, 200×150 mm	6		connected_duct_segment	(none)	matched_with_assumption	3	0
ventilation	duct_fitting	tee	IfcFlowFitting	connected 150×150 mm	4		connected_duct_segment	(none)	matched_with_assumption	2	0
ventilation	duct_fitting	transition	IfcFlowFitting	connected 150×150, 300×300, 200×150 mm	4		connected_duct_segment	(none)	matched_with_assumption	2	0
ventilation	duct_fitting	transition	IfcFlowFitting	connected 200×150, 300×300, 150×150 mm	4		connected_duct_segment	(none)	matched_with_assumption	2	0
ventilation	duct_fitting	tee	IfcFlowFitting	connected 200×150, Ø150 mm	4		connected_duct_segment	(none)	matched_with_assumption	3	0
ventilation	duct_fitting	transition	IfcFlowFitting	connected 300×150, 300×300 mm	4		connected_duct_segment	(none)	matched_with_assumption	2	0
ventilation	duct_fitting	tee	IfcFlowFitting	connected 150×150, 200×150 mm	2		connected_duct_segment	(none)	matched_with_assumption	2	0
ventilation	duct_fitting	tee	IfcFlowFitting	connected 150×150, 300×150 mm	2		connected_duct_segment	(none)	matched_with_assumption	2	0
ventilation	duct_fitting	tee	IfcFlowFitting	connected 150×150, 300×150, 200×150 mm	2		connected_duct_segment	(none)	matched_with_assumption	2	0
ventilation	duct_fitting	transition	IfcFlowFitting	connected 200×150, 150×150, 300×150 mm	2		connected_duct_segment	(none)	matched_with_assumption	2	0
ventilation	duct_fitting	transition	IfcFlowFitting	connected 300×150, 200×150 mm	2		connected_duct_segment	(none)	matched_with_assumption	2	0
ventilation	duct_fitting	tee	IfcFlowFitting	connected 300×150, Ø150 mm	2		connected_duct_segment	(none)	matched_with_assumption	3	0
ventilation	duct_fitting	cross	IfcFlowFitting	connected 300×300, 150×150, 200×150 mm	2		connected_duct_segment	(none)	matched_with_assumption	2	0
ventilation	duct_fitting	cross	IfcFlowFitting	connected 300×300, 200×150, 150×150 mm	2		connected_duct_segment	(none)	matched_with_assumption	2	0

model	kind	family	IFC class	size	elements	length m	size method	shape	status	primary products	assumption notes
ventilation	duct_fitting	transition	IfcFlowFitting	connected duct unresolved	2		unresolved	(none)	unmatched	0	0
ventilation	duct_fitting	elbow	IfcFlowFitting	connected Ø150 mm	2		connected_duct_segment	(none)	matched	1	0
ventilation	duct_fitting	transition	IfcFlowFitting	connected Ø150, 300×150 mm	2		connected_duct_segment	(none)	matched_with_assumption	3	0
ventilation	duct_flex	round_flex	IfcFlowSegment	mesh minor axis 151.6 mm	20		faceted_bbox_minor_axis	round	unmatched	0	1
ventilation	duct_flex	round_flex	IfcFlowSegment	diameter unresolved	16		unresolved	round	unmatched	0	1
ventilation	duct_segment	rectangular	IfcFlowSegment	200.0 × 150.0 mm	46	43.655	port_axis_against_extrusion	rectangular	matched_with_assumption	2	0
ventilation	duct_segment	rectangular	IfcFlowSegment	150.0 × 150.0 mm	30	31.835	port_axis_against_extrusion	rectangular	matched_with_assumption	2	0
ventilation	duct_segment	rectangular	IfcFlowSegment	300.0 × 300.0 mm	22	21.939	port_axis_against_extrusion	rectangular	matched_with_assumption	2	0
ventilation	duct_segment	rectangular	IfcFlowSegment	300.0 × 150.0 mm	18	11.051	port_axis_against_extrusion	rectangular	matched_with_assumption	2	0
ventilation	duct_segment	round	IfcFlowSegment	OD 150.0 mm	10	9.896	circle_profile	round	matched	1	0
heating_and_cooling	pipe_accessory	(none)	IfcFlowController	OD 19.05 mm	4		connected_segment_profile	(none)	matched	1	0
heating_and_cooling	pipe_fitting	elbow	IfcFlowFitting	OD 9.52 mm	37		connected_segment_profile	(none)	matched	1	1
heating_and_cooling	pipe_fitting	transition	IfcFlowFitting	OD 9.52 / 12.7 mm	30		connected_segment_profile	(none)	matched	2	0
heating_and_cooling	pipe_fitting	elbow	IfcFlowFitting	OD 12.7 mm	22		connected_segment_profile	(none)	matched	1	1
heating_and_cooling	pipe_fitting	tee	IfcFlowFitting	OD 9.52 / 12.7 mm	20		connected_segment_profile	(none)	matched	2	1
heating_and_cooling	pipe_fitting	elbow	IfcFlowTerminal	OD 9.52 mm	12		connected_segment_profile	(none)	matched	1	0
heating_and_cooling	pipe_fitting	transition	IfcFlowFitting	OD 12.7 / 19.05 mm	8		connected_segment_profile	(none)	matched	2	0

model	kind	family	IFC class	size	elements	length m	size method	shape	status	primary products	assumption notes
heating_and_cooling	pipe_fitting	transition	IfcFlowFitting	OD 19.05 mm	8		connected_segment_profile	(none)	matched	1	0
heating_and_cooling	pipe_fitting	elbow	IfcFlowFitting	OD 19.05 mm	8		connected_segment_profile	(none)	matched	1	1
heating_and_cooling	pipe_fitting	tee	IfcFlowFitting	OD 12.7 / 19.05 mm	4		connected_segment_profile	(none)	matched	2	1
heating_and_cooling	pipe_fitting	tee	IfcFlowFitting	OD 9.52 / 12.7 mm	2		connected_segment_profile	(none)	matched	2	1
heating_and_cooling	pipe_fitting	tee	IfcFlowTerminal	OD 9.52 / 12.7 mm	2		connected_segment_profile	(none)	matched	2	0
heating_and_cooling	pipe_segment	(none)	IfcFlowSegment	OD 9.52 mm	81	130.103	circle_profile	round	matched	1	1
heating_and_cooling	pipe_segment	(none)	IfcFlowSegment	OD 12.7 mm	50	60.046	circle_profile	round	matched	1	1
heating_and_cooling	pipe_segment	(none)	IfcFlowSegment	OD 19.05 mm	20	5.118	circle_profile	round	matched	1	1
heating_and_cooling	radiator	panel	IfcFlowTerminal	bbox 100 × 350 × 1000 mm	16		world_bbox	(none)	matched_with_assumption	1	0
heating_and_cooling	stove	(none)	IfcEnergyConversionDevice	bbox 350 × 500 × 1000 mm	2		world_bbox	(none)	unmatched	0	0

Architecture storeys

name	elevation m
T/FDN	-1.25
Level 1	0
Level 2	3.1
Roof	6

series	n	sum	min	max	range	mean	median	sample stdev	q1	q3
storey elevation, m	4	7.85	-1.25	6	7.25	1.962	1.55	3.254	-0.312	3.825

Architecture slabs and roofs

type	predefined type	thickness mm	z min m	z max m	count
Floor:127mm Slab on Grade	FLOOR	127	-0.127	0	2
Floor:Residential - Wood Joist with Subflooring	FLOOR	305	2.795	3.1	2
Floor:Finish Floor - Wood	FLOOR	19	3.1	3.119	6
Floor:Finish Floor - Ceramic Tile	FLOOR	13	3.1	3.113	2
Floor:Finish Floor - Wood	FLOOR	19	0	0.019	2
Floor:Finish Floor - Ceramic Tile	FLOOR	13	0	0.013	4
Basic Roof:Live Roof over Wood Joist Flat Roof	ROOF	457	6	6.457	1
Floor:150mm Exterior Slab on Grade	FLOOR	150	-0.137	0.013	2

series	n	sum	min	max	range	mean	median	sample stdev	q1	q3
slabs thickness, one type row, mm	8	1103	13	457	444	137.875	73	164.528	17.5	188.75
slabs thickness, one instance, mm	21	1851	13	457	444	88.143	19	125.081	13	127

Architecture ceilings

type	predefined type	thickness mm	z min m	z max m	count
Compound Ceiling:Gypsum Board	CEILING	57	2.5	2.557	4
Compound Ceiling:Gypsum Board	CEILING	57	5.6	5.657	8

series	n	sum	min	max	range	mean	median	sample stdev	q1	q3
ceilings thickness, one type row, mm	2	114	57	57	0	57	57	0		
ceilings thickness, one instance, mm	12	684	57	57	0	57	57	0	57	57

Architecture beams

type	predefined type	thickness mm	z min m	z max m	count
M_W-Wide Flange:W310X60	(none)	303	2.797	3.1	2
M_W-Wide Flange:W410X60	(none)	407	2.693	3.1	2
M_W-Wide Flange:W410X60	(none)	407	5.593	6	4

series	n	sum	min	max	range	mean	median	sample stdev	q1	q3
beams thickness, one type row, mm	3	1117	303	407	104	372.333	407	60.044		
beams thickness, one instance, mm	8	3048	303	407	104	381	407	48.143	381	407

Ceiling to structure gaps

ceiling z max m	ceiling thickness mm	overhead	overhead z min m	gap mm	ceilings
2.557	57	Floor:Residential - Wood Joist with Subflooring	2.795	238	4
5.657	57	Basic Roof:Live Roof over Wood Joist Flat Roof	6	343	8

series	n	sum	min	max	range	mean	median	sample stdev	q1	q3
clear gap, one ceiling instance, mm	12	3696	238	343	105	308	343	51.698	238	343
ceiling thickness in the gap table, one instance, mm	12	684	57	57	0	57	57	0	57	57

The floor and roof build-ups above the rooms use dimensional lumber and plywood, not a concrete soffit.

Ceilings are compound gypsum board on a metal-stud layer (57 mm in the material layer set), so a concrete ceiling anchor does not match this model's soffit.

Concrete in this architecture model is the slab on grade and the foundation walls, not the ceiling the MEP hangers would reach.

Sources

series	n	sum	min	max	range	mean	median	sample stdev	q1	q3
sources cited by one product	23	31	1	2	1	1.348	1	0.487	1	2

accessed	citations	share of citations	cumulative share
2026-10-18	36	100.0%	100.0%

source host	citations	share of citations	cumulative share
www.meinhausshop.de	5	13.9%	13.9%
www.dibt.de	4	11.1%	25.0%
www.muepro.de	3	8.3%	33.3%
1001krep.ru	3	8.3%	41.7%
www.sikla.de	3	8.3%	50.0%
heizung-billiger.de	2	5.6%	55.6%
assets.hilti.com	2	5.6%	61.1%
www.fischer.de	2	5.6%	66.7%
fiproductmedia.azureedge.net	2	5.6%	72.2%
www.fischer-international.com	1	2.8%	75.0%
www.unielektro.de	1	2.8%	77.8%
shop.famo24.de	1	2.8%	80.6%
www.sikla.ch	1	2.8%	83.3%
www.hilti.de	1	2.8%	86.1%
www.beijerref.de	1	2.8%	88.9%
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cdn.standards.iteh.ai	1	2.8%	97.2%
doi.org	1	2.8%	100.0%

title	url	citations	host
fischer datasheet Rohrschelle FRS Plus	https://www.meinhausshop.de/media/pdf/94_046_22--DBL-01-de.pdf	4	www.meinhausshop.de
Distributor reproduction of the Hilti MP-MI load table	https://1001krep.ru/f/mp-mi.pdf	3	1001krep.ru
ETA-18/0130 Hilti MP-MI pipe clamps	https://www.dibt.de/pdf_storage/2024/ETA-18!0130(8.06.02-122!21).pdf	3	www.dibt.de

title	url	citations	host
Müpro OPTIMAL Rohrschelle article table	https://www.muepro.de/Produkte/Befestigungstechnik/Rohrschellen/OPTIMAL.html	3	www.muepro.de
fischer EA II permissible-load table	https://fiproductmedia.azureedge.net/media/Marketing%20Materials/Sales%20Materials/VKU_EAII_F_SDE_AIP_V2.pdf	2	fiproductmedia.azureedge.net
fischer SHK catalogue, Einschlaganker EA II article table	https://www.fischer.de/-/media/fixing-systems/rebrush/fide/service/infos-und-updates/informationsmaterial/documents/kataloge/katalog-sanitaer-v1.pdf	2	www.fischer.de
Sikla Lüftungswinkel LUW	https://www.sikla.de/startseite-131533/produkte-131535/e-katalog-131895/siconnect-3832/lueftung-10505/lueftungswinkel-luw-4160	2	www.sikla.de
Hilti manual, medium-light installation systems, M8 threaded rod table	https://assets.hilti.com/api/public/content/5021068-original?v=Manuale+Sistemi+installazione+medio+leggere.pdf	1	assets.hilti.com
Hilti manual, medium-light installation systems, M10 threaded rod table	https://assets.hilti.com/api/public/content/5021068-original?v=Manuale+Sistemi+installazione+medio+leggere.pdf	1	assets.hilti.com
EN 12236:2002 preview (SIST EN 12236:2002)	https://cdn.standards.iteh.ai/samples/6392/f9a3d57c81574acb8ee255867c6daaca/SIST-EN-12236-2002.pdf	1	cdn.standards.iteh.ai
DIN EN 12236:2002-04 bibliographic record	https://doi.org/10.31030/9199301	1	doi.org
Kermi Wandkonsolen-Set ZB02870003 reseller technical text	https://heizung-billiger.de/44039-kermi-wandkonsolen-set-variabel-mit-tiefenverstellung-wa-60-100-mm-weiss-kermi-zb02870003-4037486266503.html	1	heizung-billiger.de
Walraven BISMAT 2000 100–105 mm reseller listing (3403105)	https://heizung-billiger.de/727280-walraven-bismat-rohrschelle-2000-100-105mm-m8-m10-zweischraubenschelle-epdm-stahl-elektrolytisch-verzinkt-walraven-3403105-8712993004126.html	1	heizung-billiger.de
Lindab Ventilation, Upphängning av ventilationssystem	https://itsolution.lindab.com/lindabwebproductsdoc/pdf/documentation/ads/se/mounting/upphangning-ventsystem.pdf	1	itsolution.lindab.com
Walraven BIS KSB 2 100–105 mm reseller listing (Werksnummer 3396105)	https://shop.famo24.de/bis-zweischraubenschelle-ksb-2-100-105-mm-m8-10-ev-epdm/14810134	1	shop.famo24.de
IBP technical catalogue, section 4.5 Befestigungsabstände	https://storage.ua.prom.st/657649_supersan_ibp_tk_all.pdf	1	storage.ua.prom.st
Conex installation notes, section 3.4 Befestigungsabstände	https://www.beijerref.de/media/ed/5c/fb/1758265815/conex-allgemeine-installationshinweise-ast-2241085.pdf	1	www.beijerref.de
ETA-18/0119 Hilti MQ installation channels	https://www.dibt.de/pdf_storage/2024/ETA-18!0119(8.06.02-121!21).pdf	1	www.dibt.de
fischer Pipe clamp FRS-L Universal	https://www.fischer-international.com/en/products/installation-systems/pipe-clamps/two-screw-pipe-clamps/pipe-clamp-frs-l-universal	1	www.fischer-international.com
Hilti MQ-41/3 product page	https://www.hilti.de/c/CLS_MODULAR_SUPPORT_SYSTEM/CLS_MODULAR_SUPPORT_PRO/r6130	1	www.hilti.de
fischer datasheet Rohrschellen FRS A2/A4	https://www.meinhausshop.de/media/pdf/meinhausshop-frs-a2-de.pdf	1	www.meinhausshop.de
Sikla Lüftungsschelle PLU 2G-N m.E.	https://www.sikla.ch/startseite-131533/produkte-131535/e-katalog-131895/siconnect-3832/lueftung-10505/lueftungsschelle-plu-2g-n-me-116704	1	www.sikla.ch
Sikla Montagetechnik, Lüftungswinkel SZ note	https://www.sikla.de/fast/600/Montagetechnik_2023-04.pdf	1	www.sikla.de
Reseller listing FISCHER 539443 FRS-L Universal 8-11	https://www.unielektro.de/FISCHER-539443-FRS-L-Universal-8-11-Rohrschelle--item-01253925	1	www.unielektro.de

Distinct source URLs: 24. Citations counting a reused URL once per product or guidance record: 36.

Assumptions stored on products

id	assumption
sikla-luw-sl-189893	The LUW datasheet describes suspension of rectangular ducts and does not publish a width or height window.
sikla-luw-sz-189902	The LUW datasheet describes the SZ angle for a wide, low rectangular duct and does not publish a width or height window.
kermi-console-zb02870003	The console is for Kermi radiators with hanging lugs 25 mm high and 2 mm thick. The duplex IFC radiator is named only Radiator and does not state a manufacturer or lugs.

Product assumption strings: 3 on 3 products.

IFC2X3 export: pipes and ducts are IfcFlowSegment, not IfcPipeSegment / IfcDuctSegment.

Segment property sets have length and roughness, not nominal diameter. Outside diameter is the circle-profile diameter.

Rectangular duct profiles mix run length into the profile. Width and height are the two edges that are not the port-to-port direction; height is the edge closer to world Z.

No insulation covering or insulation property was found on the MEP elements.

On fittings, NominalDiameter is an IfcPropertyListValue and often disagrees with the connected pipe's circle-profile diameter. Groups keep both; the profile diameter is the one a clamp has to fit.

Material entities on the MEP models are unnamed.